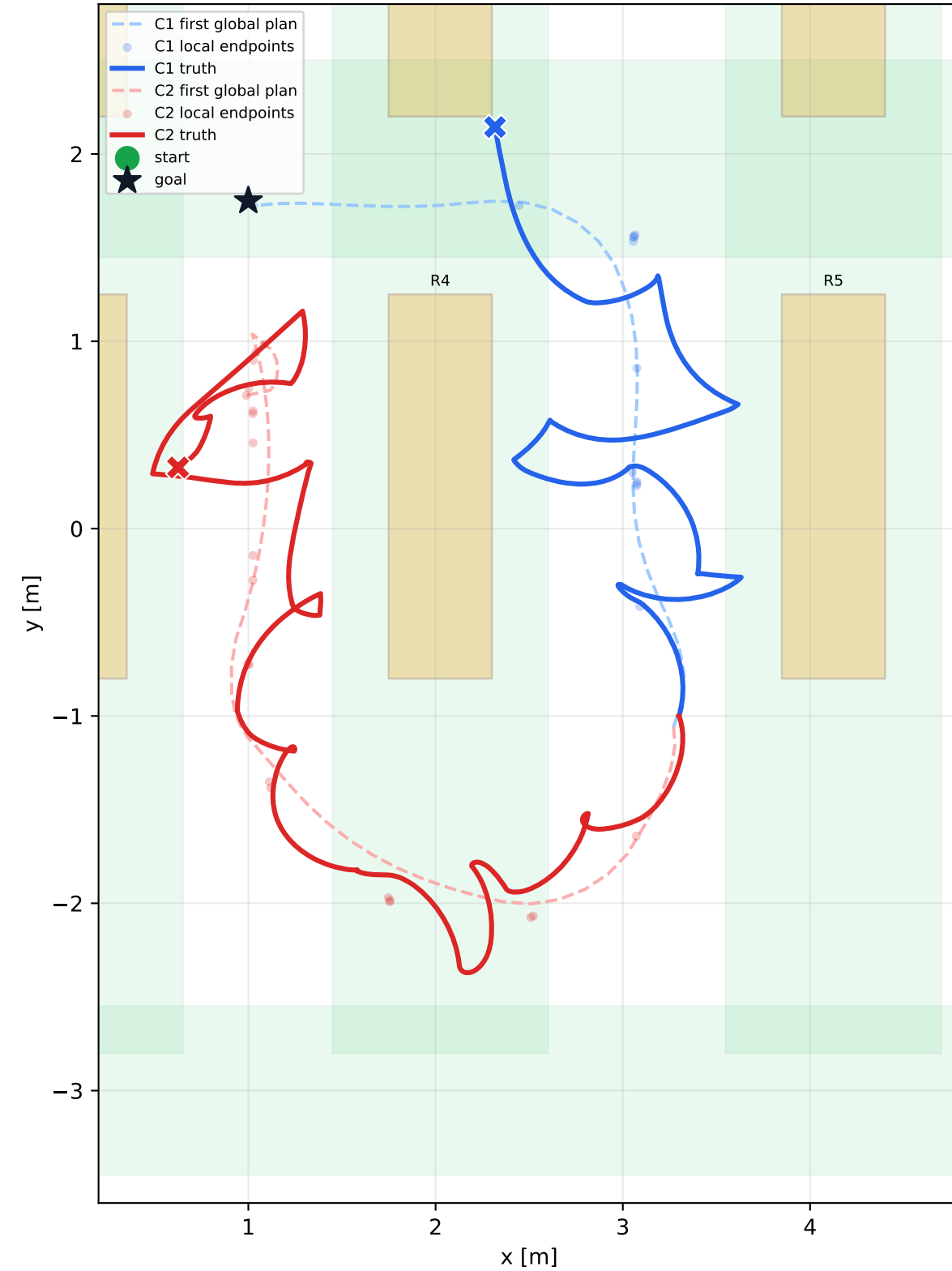


F45 - Tracking yaw + local solver timing diagnostic

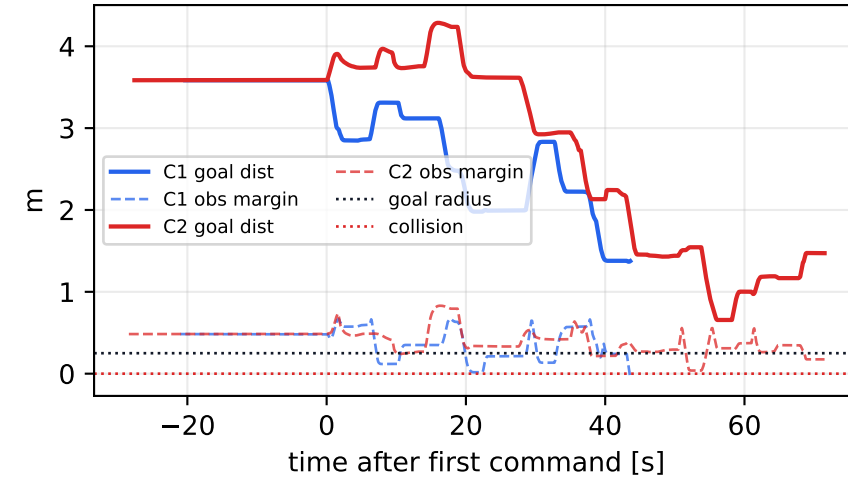
R4

R5

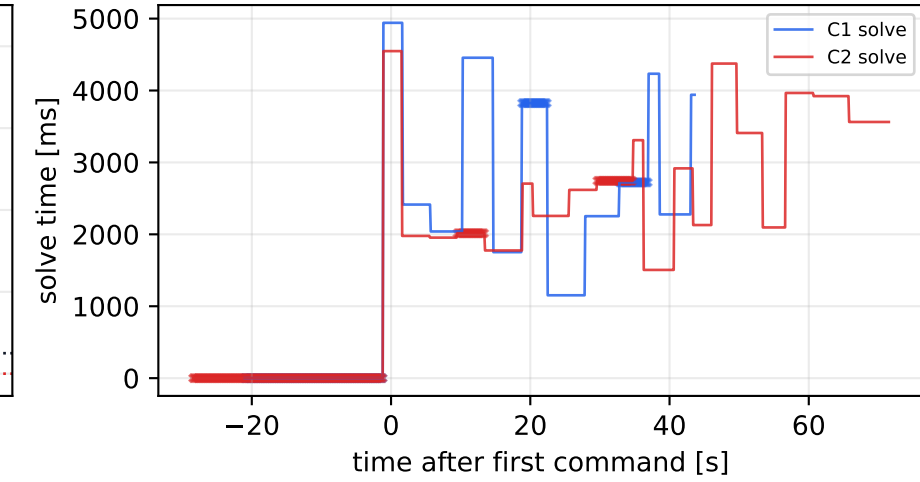
(a) First route, local endpoints, executed trajectory



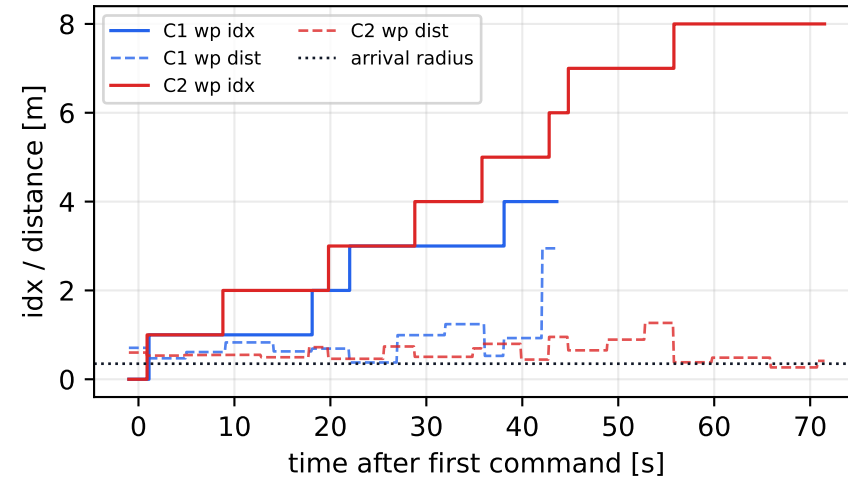
(b) Goal progress and obstacle margin



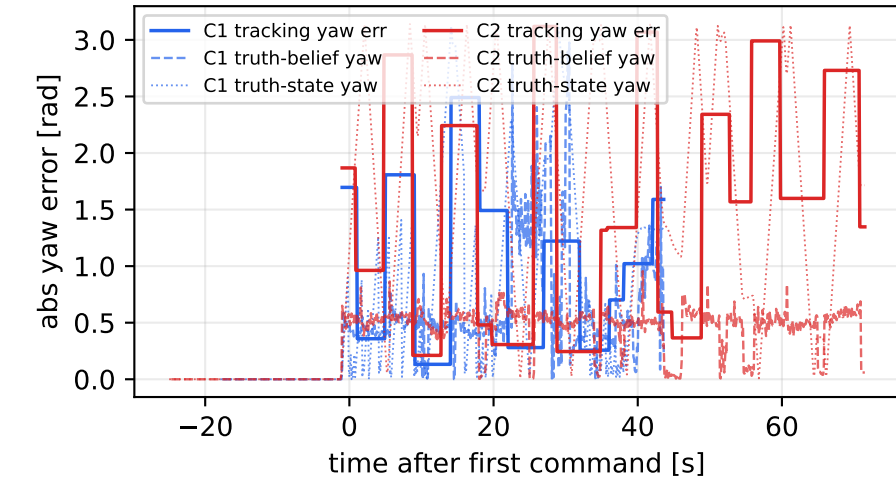
(c) Local solve time (x = non-success)



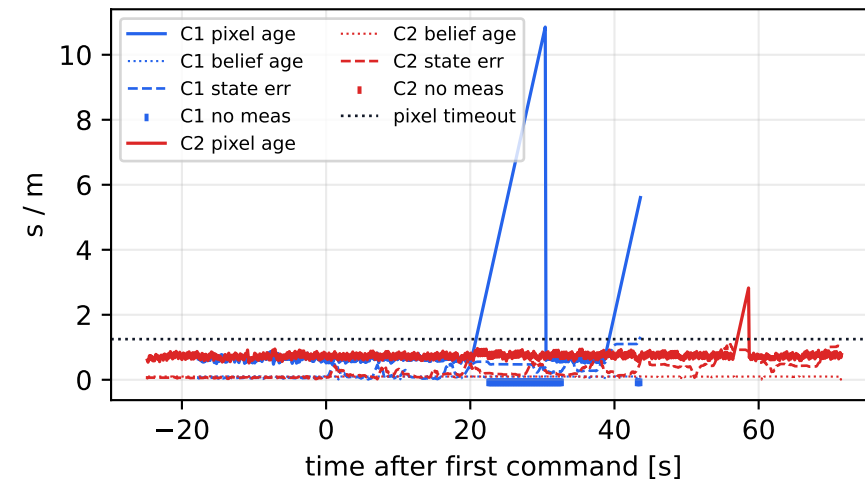
(d) Waypoint progress



(e) Heading/yaw errors



(f) Perception freshness and localization



F45 controlled tracking-yaw smoke

C1: collision, path=6.94m, min_goal=1.37m
err=0.282m, min obs=-0.055m
solve μ /p90=3000/4433 ms, 62 ms/eval
maxiter/non-success rows=12%
yaw track μ /p90=1.02/1.81 rad
pixel age p90/max=5.19/10.85s

C2: stuck, path=10.23m, min_goal=0.66m
err=0.290m, min obs=0.038m
solve μ /p90=2831/4048 ms, 66 ms/eval
maxiter/non-success rows=9%
yaw track μ /p90=1.56/2.99 rad
pixel age p90/max=0.86/2.83s

Interpretation:

1. Spawn yaw is clean; yaw problems are runtime/tracking, not init.
2. Global plans still choose the expected route families.
3. Local H15/maxiter20 is too budget-starved: frequent maxiter hits.
4. C1 additionally suffers stale camera belief before collision.
5. C2 localizes better but still collides: local tracking/clearance remains failing